Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor | Unified Mentor Project

Executive Summary

This project analyzes 10,194 shipment/order-line records for Nassau Candy Distributor to understand factory-to-customer route structure, shipment volume, ship-mode usage, geographic distribution, and commercial performance. Product names were mapped to five factories using the mapping supplied in the project brief. The dataset contains 8,549 unique orders, 15 products, 59 state/province labels, 4 regions, and 4 shipping modes.

A key data-quality issue was identified: parsed order dates range from 2024-01-02 to 2025-12-31, while ship dates range from 2026-06-30 to 2030-06-28. The resulting lead time ranges from 904 to 1642 days, with a median of 1274 days. These values are not credible as normal delivery durations. Therefore, this report does not claim that any route is genuinely fastest or slowest, and delay-frequency KPIs should remain provisional until source dates are verified.

1. Introduction

Shipping route efficiency affects delivery reliability, customer experience, and logistics cost. This analysis links each product to its originating factory and then groups records by factory-to-customer state and factory-to-region. It provides a first-pass view of route volumes and ship-mode patterns while clearly identifying limitations in the available date fields.

2. Problem Statement

The organization needs visibility into route volumes, geographic concentrations, shipping-method usage, and potential operational bottlenecks. However, route-level delivery-time conclusions depend on accurate order and ship dates. The supplied file contains a major date inconsistency, which limits valid lead-time benchmarking.

3. Objectives

• Map each product to the appropriate factory.

• Construct factory-to-state and factory-to-region routes.

• Summarize route volume, orders, units, sales, and gross profit.

• Compare shipment records across ship modes and regions.

• Audit date quality before using lead time or delay frequency.

• Deliver a filterable Streamlit dashboard for exploration.

4. Dataset and Data Preparation

The original dataset contains 10,194 rows and 18 source columns. Key fields include Order ID, Order Date, Ship Date, Ship Mode, City, State/Province, Region, Product Name, Sales, Units, Gross Profit, and Cost.

Preparation steps:

1. Read the CSV and inspect its schema.

2. Parse dates using the day-month-year format (`%d-%m-%Y`).

3. Map Product Name to Factory using the supplied product-factory table.

4. Create `Factory → State` and `Factory → Region` route labels.

5. Calculate raw shipping lead time as Ship Date minus Order Date.

6. Add a date-quality flag to identify missing/invalid, negative, plausible (0–30 days), and suspicious (>30 days) lead times.

7. Preserve the original source file and save derived columns in a separate working CSV.

5. Initial Data Audit

6. Date Quality Finding

Order dates: **2024-01-02 to 2025-12-31**

Ship dates: **2026-06-30 to 2030-06-28**

Calculated lead time: **minimum 904 days, median 1274 days, maximum 1642 days**.

Counts:

• Lead times from 0 to 30 days: 0

• Lead times above 30 days: 10,194

• Negative lead times: 0

• Missing/invalid calculated lead times: 0

The date ranges and lead-time magnitudes indicate a source-data inconsistency. A plausible delivery duration must not be invented by subtracting an arbitrary number of years or days. Recommended action: confirm the intended date format and date-generation process with the dataset provider, then rerun lead-time and delay analysis.

7. Analytical Methodology

Route construction: Product Name → Factory → Customer State/Region.

Volume analysis: Count records and unique orders by route.

Commercial analysis: Sum Sales, Gross Profit, and Units by route, factory, state, region, and ship mode.

Ship-mode analysis: Compare shipment counts and commercial metrics across Standard Class, Second Class, First Class, and Same Day.

Lead-time validation: Calculate date differences, then flag suspicious values before interpreting delivery performance.

8. Dashboard Design

The Streamlit app includes:

• KPI cards for shipment records, unique orders, sales, and gross profit.

• Filters for order-date range, region, state, ship mode, and factory.

• Top factory-to-state routes by record count.

• Sales contribution by factory.

• Ship-mode shipment-volume and gross-profit comparisons.

• Geographic volume visualization by region and state.

• Route drill-down with order-level records.

• Date-quality audit and an explicit limitation warning.

• CSV download of the filtered records.

9. Findings and Interpretation

1. The product-to-factory mapping covers all product records in the supplied file; no unmapped products remain after applying the provided mapping.

2. The data supports descriptive analysis of route volume, geographic distribution, ship-mode usage, sales, units, and gross profit.

3. The date-derived lead times are not credible for normal shipping performance and should not be used to rank routes or calculate delay rates until corrected at source.

4. High-volume routes can be prioritized for operational review, but volume alone does not prove congestion or poor delivery performance.

5. A genuine cost-time trade-off cannot be established from ship mode alone without trustworthy delivery times and a clear shipping-cost field.

10. Recommendations

• Ask the dataset provider to verify Order Date and Ship Date values and formatting.

• Retain the original CSV and record any future corrections in a reproducible cleaning script.

• Once dates are validated, calculate average and median lead time, lead-time variability, and the percentage exceeding an agreed threshold.

• Review high-volume factory-to-state routes first for capacity planning.

• Compare ship modes only after date validity and shipping cost data are confirmed.

• Track route volume and lead-time KPIs over time in the dashboard.

11. Limitations

• The dataset does not include a direct customer-coordinate table, and only factory coordinates are supplied; a true route map requires customer/state coordinates or a geospatial lookup.

• The provided data does not establish a validated delivery-time threshold.

• The date anomaly prevents defensible route-speed ranking, delay frequency, and delivery-time/cost trade-off conclusions.

• Shipment records may represent order lines rather than one unique physical shipment; counts should therefore be described as records unless shipment IDs are available.

12. Conclusion

The project creates a repeatable foundation for factory-to-customer logistics analysis by mapping products to factories, defining routes, profiling shipment volume and commercial metrics, and providing a Streamlit dashboard. Its most important quality finding is that the supplied dates do not support reliable lead-time analysis in their current form. Verifying those dates is necessary before the organization can responsibly identify the fastest routes, quantify delays, or make delivery-performance claims.

Appendix: Factory Coordinates